

STAT 230A Final Report

Evalynna Ong

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1 Introduction

San Francisco is characterized by its multitude of neighborhoods, likened to distinct microcosms of varying culture, architecture, and even climate. Major differences in socioeconomic status exist in tandem with these neighborhood quirks throughout the city, with San Francisco having one of the most extreme wealth gaps in the nation. There are many studies on the correlation between socioeconomic status and health [1], [2], [3], [4]. Evans and Kantrowitz cite environmental risk exposure as a contributing factor [1]. For the city to best provide equitable support to its people, it is important to understand the impact of these health hazards and the resulting needs of the impacted communities. This project evaluates data on requests for emergency medical services and pollution measurements by location. The primary question being whether communities disproportionately burdened by multiple sources of pollution experience higher need for emergency medical services. Specifically, this project aims to evaluate emergency call volume as a function of the pollution data and perform statistical inference on the coefficients of contributing pollutants.

2 Data and Domain Problem

The San Francisco Department of Emergency Management records data from their dispatch system. As 911 calls come in, incident information (including the longitude and latitude of the incident) is automatically recorded. The outcome variable for this project will be a count of calls per week by location over the course of 2024.

The covariates used to model call volume come from the Office of Environmental Health Hazard Assessment (OEHHA). Communities are identified by their census location and concentrations of pollutants, reports on health issues, and socioeconomic scores are reported for each year. This project will evaluate 198 census locations from the city of San Francisco ($n=198$) for 2024. Because the various hazard covariates are recorded in different measures, they are normalized.

The San Francisco Department of Emergency Management does not organize their data by census locations. However, each incident contains longitude and latitude data of where the call came from. OEHHA includes longitude and latitude data for the centroid of each census location and so each emergency incident is given an associated census tract based on its nearest location.

The primary model for analysis considers census tract as the observations with a secondary evaluation of zipcode-by-week for comparison. Suspecting potential seasonal trends in call volume, averages of call counts per week were taken over three month periods and attributed to each observation for seasonal evaluation. One limitation on this data is the static nature of the OEHHA dataset. The pollutant covariates do not change over time and instead represent an aggregate version of pollution levels for 2024. As a result, the seasonal comparison is treated as a stability analysis, evaluating whether the same cross-sectional relationships hold across different outcome windows. Rather than a longitudinal analysis of how changes in pollution affect call volume.

3 Regression Model

Because the outcome variable explores discrete count data, Poisson regression was initially considered. However, exploratory data analysis revealed that the variance of weekly call counts exceeded the mean across

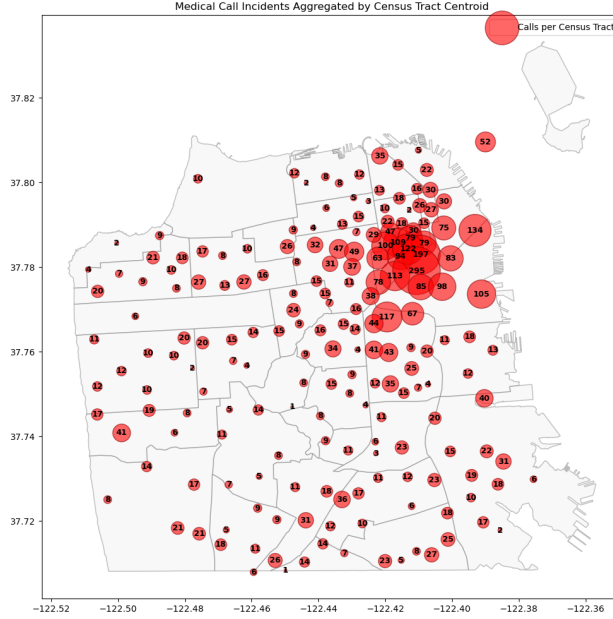


Figure 1: Average Medical Call Incidents per census tract per week for August 2024 - October 2024.

census tracts. As a result, Negative Binomial regression was selected to model the data and account for this overdispersion.

An initial model was evaluated with all 22 OEHHHA hazard covariates to establish a baseline for the relationship between pollution and environmental burden with EMS call volume. The R^2 was 0.175, indicating the covariates collectively explain approximately 17.5% of the variance in call volume, with the likelihood ratio test confirming the model was significantly better than a null intercept-only model ($p < 10^{-40}$). From this initial model, poverty was the most dominant predictor and hazardous waste was the most dominant pollution indicator. Some counterintuitive results, like the opposite sign for unemployment as poverty level, indicated some collinearity across covariates. A correlation heatmap and VIF evaluation revealed significant collinearity (see Additional Figures).

To address this, Elastic Net regularization was applied via the statsmodels GLM with Negative Binomial family. Hyperparameters α and L_1 weight were selected through 5-fold cross-validation, minimizing mean absolute error. The optimal parameters ($\alpha = 0.01$, L_1 weight = 0.2) produced a model that zeroed out drinking water contamination, toxic release, traffic, solid waste, education, linguistic isolation, cardiovascular disease, and unemployment. The 12 remaining variables were then used to refit an unpenalized Negative Binomial to obtain valid standard errors and p-values for inference.

4 Discussion

The evidence partially supports the hypothesis that communities disproportionately burdened by pollution experience higher EMS demand. The analysis identifies hazardous waste proximity, poverty, asthma prevalence, and housing burden as the most robust positive predictors of call volume. A one standard deviation increase in hazardous waste burden is associated with approximately a 25% increase in expected weekly EMS calls ($\exp(0.237) \approx 1.27$). Poverty shows a similar magnitude ($\exp(0.158) \approx 1.17$). The consistency of these findings across seasons, with covariates being static, indicates the same communities experience disproportionate emergency burden year-round. This suggests a structural rather than seasonal phenomenon.

Pollution burden and poverty are deeply correlated in San Francisco. Low-income communities are disproportionately located near industrial facilities, hazardous waste sites, and high-traffic corridors. The model estimates their independent effects. However, the magnitude of pollution effects is likely understated because part of what appears as a poverty effect is the cumulative consequence of environmental exposure.

These findings are best interpreted as evidence that EMS demand is driven by a combination of environmental and socioeconomic burden.

For decision-makers, these results suggest that EMS resource allocation in San Francisco should prioritize census tracts with high hazardous waste burden and high poverty concentration. The stability of these findings across all four seasonal windows indicates resource planning does not need to be seasonally adjusted for these communities.

Comparing the census tract results to the zip code aggregation reveals important scale-dependent findings. Several covariates flip sign at the zip code level, most notably hazardous waste and cleanup sites becoming negative, and asthma becoming negative and losing significance. Zip codes are likely too broad of a geographic unit to capture the neighborhood-level variation that drives the relationships of interest. When census tracts with very different pollution burdens are averaged into a single zip code, the within-zip variation that identifies these effects is lost. This comparison reinforces the census tract as the appropriate primary unit of analysis.

4.0.1 Limitations

Several limitations constrain the scope of this analysis. The static nature of the OEHHA covariate data allows the model to identify relationships between pollution burden and EMS demand, but it cannot address how temporal changes in pollution affect call volume. Consequently, the seasonal comparison functions as a stability analysis rather than a longitudinal one.

Additionally, the outcome variable—the average weekly call count over a three-month period—is a continuous approximation of a count outcome. Furthermore, data from additional years is required to truly evaluate the impact of these covariates. Finally, the use of averaged and subsequently aggregated data introduces smoothing, meaning these results may be subject to significant change with more granular data.

Table 1: Negative Binomial Regression Coefficients Across Model Specifications

(lr)2-5 Variable	Reduced Model: Census Tract (Clustered SE)				Zip Code	Full Model
	Aug–Oct	Nov–Jan	Feb–Apr	May–Jul		
<i>Pollution Covariates</i>						
Ozone	−0.216	−0.167	−0.216	−0.190	−0.907	−0.194
Diesel PM	0.088	0.084	0.100	0.094	−0.002	0.040
Lead	−0.044	−0.070	−0.004	−0.053	−0.354	−0.061
Pesticides	−0.117	−0.119	−0.143	−0.125	−0.382	−0.099
Cleanup Sites	0.145	0.122	0.138	0.118	−0.022	0.142
Groundwater Threats	−0.193	−0.209	−0.198	−0.210	−0.609	−0.199
Hazardous Waste	0.237	0.254	0.254	0.251	−0.207	0.227
Impaired Water Bodies	−0.162	−0.163	−0.157	−0.142	−0.650	−0.162
<i>Health Indicators</i>						
Asthma	0.262	0.278	0.253	0.242	−0.497	0.176
Low Birth Weight	0.084	0.066	0.068	0.072	−0.174	0.100
<i>Socioeconomic Covariates</i>						
Poverty	0.158	0.160	0.146	0.181	0.475	0.266
Housing Burden	0.091	0.095	0.115	0.089	0.669	0.108
<i>Model Fit</i>						
Pseudo R^2	0.171	0.164	0.168	0.161	—	0.175
Log-Likelihood	−624.87	−631.23	−623.82	−630.48	—	−618.77
$\hat{\alpha}$	0.147	0.152	0.155	0.169	0.024	0.139
N	182	182	182	182	~30	182

Notes: All covariates are z-score standardized. The reduced model retains the 12 variables surviving Elastic Net cross-validation (best parameters: $\alpha = 0.01$, L_1 weight = 0.2, selected by 5-fold CV minimizing MAE). Seasonal models use identical covariate sets with standard errors clustered by census tract. Zip code model uses the full covariate set aggregated by zip code (pollution covariates averaged, call counts summed across constituent tracts; $N \approx 30$). Full model uses all 22 covariates without regularization, standard errors nonrobust.

5 Additional Work

Several analyses were conducted to evaluate the robustness of the primary findings.

5.0.1 Seasonal stability

The reduced model was refit separately for each of four three-month seasonal windows (Aug-Oct, Nov-Jan, Fe-Apr, May-Jul) with standard errors clustered by census tract. The distribution of call volume across census tracts was evaluated for seasonal differences (see Figure 3.), which revealed no statistically significant differences across seasons. Coefficient estimates for the key predictors — hazardous waste, groundwater threats, impaired water bodies, asthma, cleanup sites, poverty, and ozone — were consistent in sign across all four seasonal windows. This consistency provides strong evidence that the cross-sectional relationships are structural rather than seasonal artifacts.

5.0.2 Single week evaluation

To assess the effect of outcome aggregation, the model was additionally fit on the raw call count for a single representative week. The dispersion parameter increased substantially ($\hat{\alpha} = 0.300$ vs. 0.147 for the seasonal average), confirming that averaging over 13 weeks mechanically deflates the apparent dispersion. The pseudo- R^2 fell from 0.171 to 0.118, reflecting the noise of a single week as an outcome.

5.0.3 Zipcode aggregation

The model was refit aggregating census tract observations to zip code level with hazard covariates averaged and call counts summed across tracts. Following elastic net cross validation, the coefficient estimates differed significantly from the census tract-level model. Other works show that census tracts provide estimates of socioeconomic gradients “on par with those obtained with individual-level socioeconomic measures” and are otherwise unaffected by problems unique to the zipcode level data [5].

5.0.4 Hazard covariate evaluation

Vizualization of hazard covariate intensity across the San Francisco census tracts. There is clear geographic clustering and this heterogeneity is likely the main cause of collinearity and counterintuitive sign issues in this regression (Figure 5.)

6 Conclusion

This project evaluated whether communities disproportionately burdened by environmental pollution experience higher need for emergency medical services. Using Negative Binomial regression to model weekly EMS call volume, 182 San Francisco census tracts as a function of pollution, health, and socioeconomic indicators from the OEHHA CalEnviroScreen 4.0 dataset.

The dominant role of poverty alongside pollution indicators reflects how environmental hazard exposure and socioeconomic disadvantage are geographically co-located in San Francisco, and their joint effect on EMS demand exceeds what either factor predicts independently.

Not all pollution covariates behaved as hypothesized. Ozone, groundwater threats, and impaired water bodies showed negative and statistically significant associations with call volume across all specifications. These counterintuitive results are most likely explained by spatial confounding specific to San Francisco’s geography. Next steps would be to evaluate data across additional sources that impact community needs for emergency medical services.

Multiple assumptions were made throughout the data engineering and modeling processes. The most consequential assumption was that the static OEHHA covariate data is assumed to represent a stable, time-invariant pollution burden. As a result, this model only identifies cross-sectional relationships between environmental hazards and EMS demand but cannot be used to interpret how changes in exposure to environmental hazards over time impact EMS demand. The findings presented in this evaluation offer meaningful preliminary evidence that environmental and socioeconomic burden predict EMS demand in San

Francisco. However, the limitations of the static covariates and single-year outcome window mean these results should be interpreted more as exploratory than definitive. Replication of this analysis with multiple years of EMS call data and updating OEHHHA covariate data would be necessary to establish true stability of the relationship.

References

- [1] G. W. Evans and E. Kantrowitz, “Socioeconomic Status and Health: The Potential Role of Environmental Risk Exposure,” *Annu. Rev. Public Health*, vol. 23, pp. 303–331, 2002.
- [2] G. Agarwal et al., “Social factors in frequent callers: a description of isolation, poverty and quality of life in those calling emergency medical services frequently,” *BMC Public Health*, vol. 19, p. 684, 2019.
- [3] A. Härenstam et al., “Patterns of working and living conditions,” *Work Stress*, vol. 17, pp. 73–92, 2003.
- [4] O. Lundberg, “The impact of childhood living conditions on illness and mortality in adulthood,” *Soc. Sci. Med.*, vol. 36, pp. 1047–1052, 1993.
- [5] N. Krieger, “A Century of Census Tracts: Health & the Body Politic (1906–2006),” *J Urban Health*, vol. 83, no. 3, pp. 355–361, May 2006, doi: 10.1007/s11524-006-9040-y.

A Additional Figures and Tables

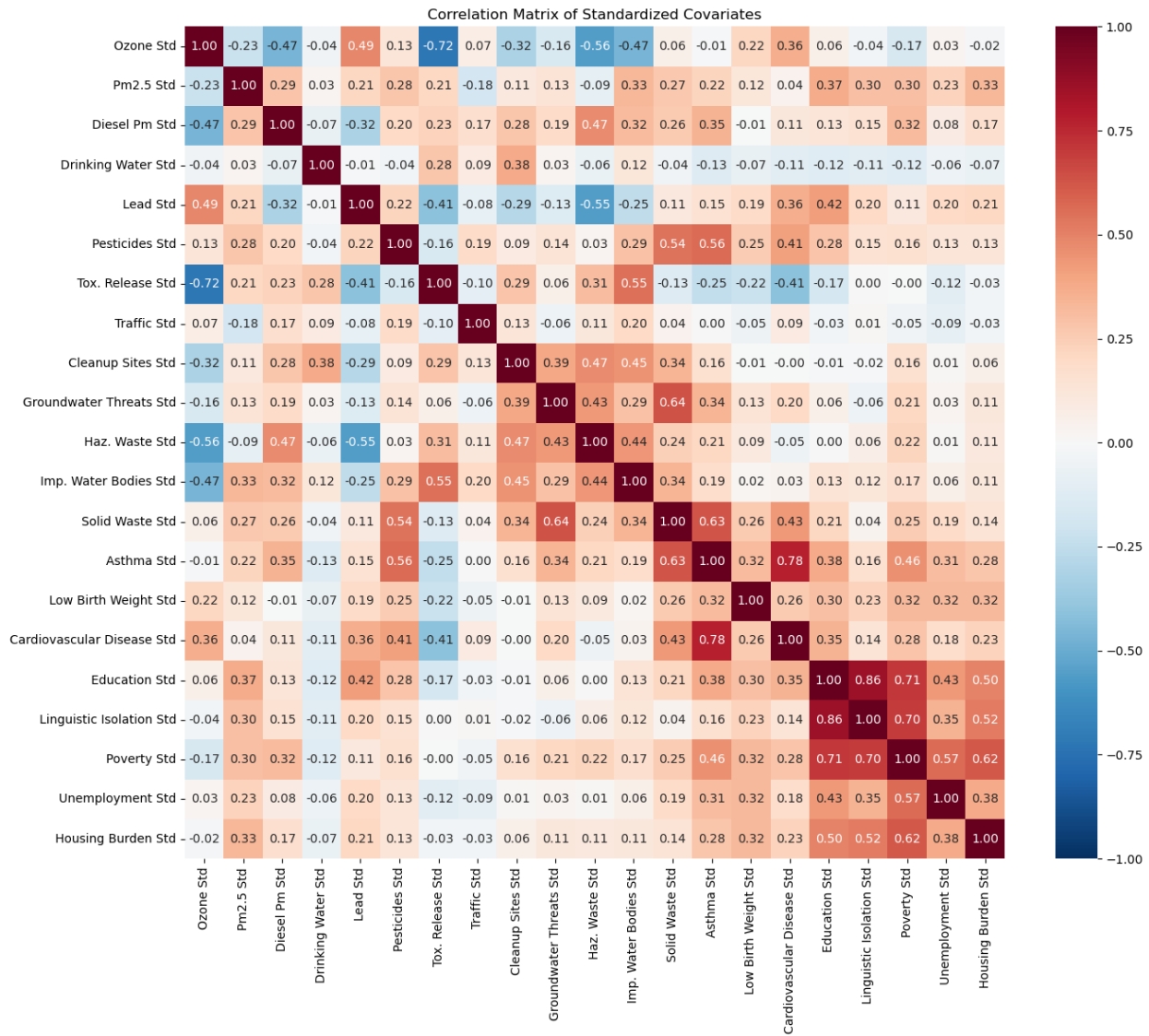


Figure 2: Correlation matrix of all 22 OEHHHA hazard covariates.

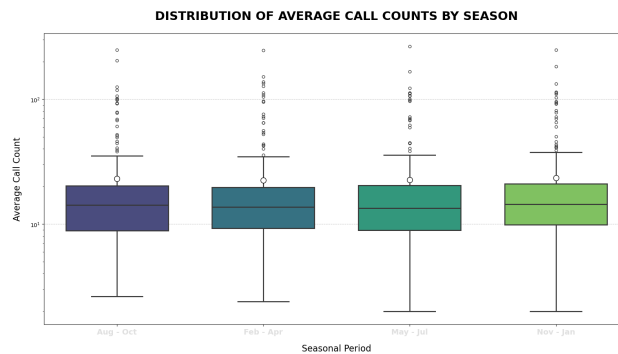


Figure 3: Distribution of average call counts per census tract by season

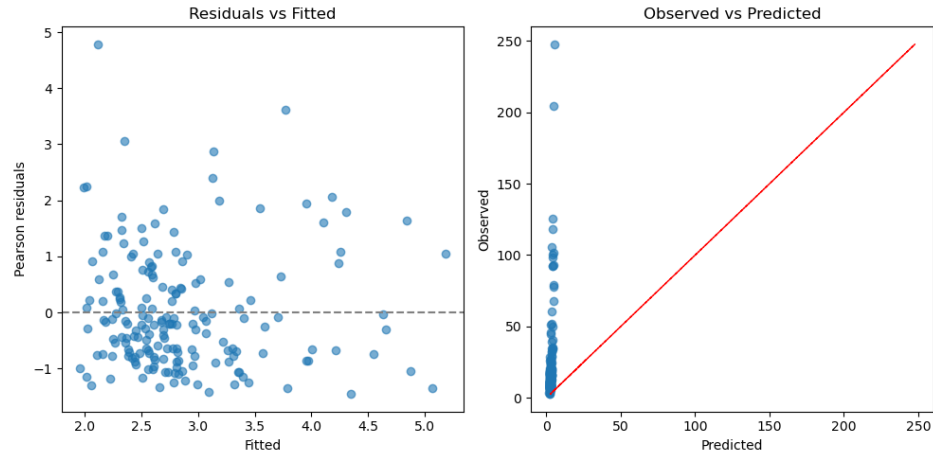


Figure 4: (Left) Residuals versus fitted (Right) Observed versus Predicted for the reduced model following Elastic Net feature selection

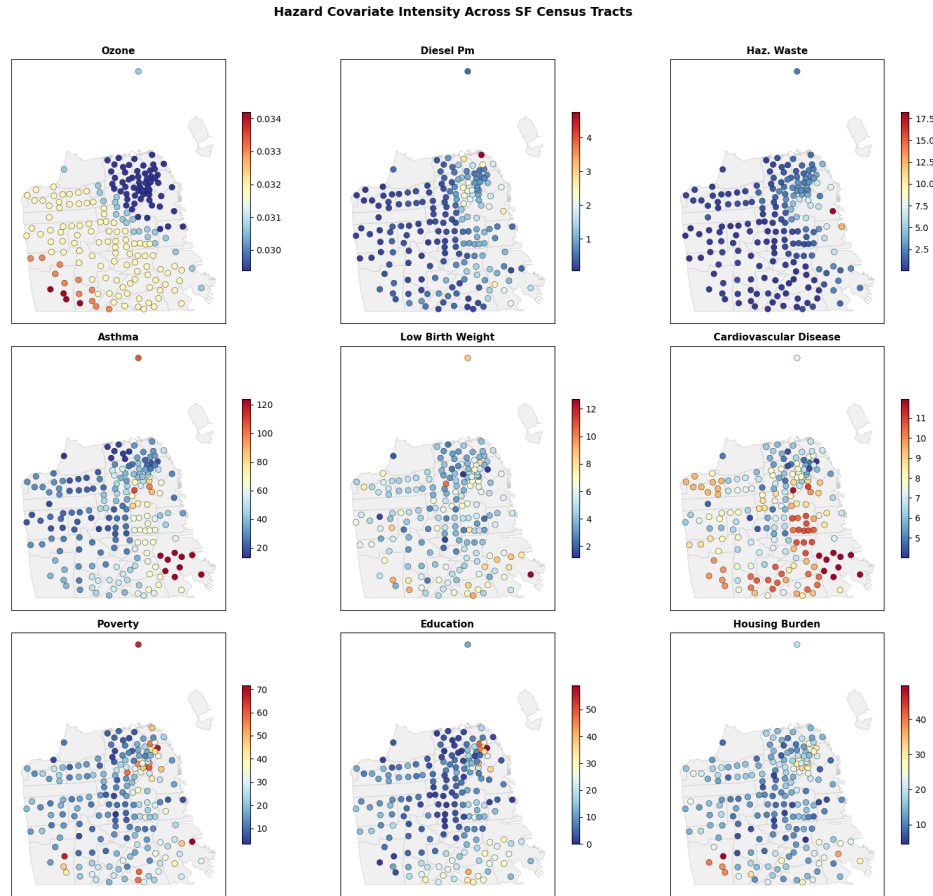


Figure 5: Hazard covariates intensity across SF census tracts for the indicators with the 9 most impactful covariates.